

HW 2

Problem 4.2

a. To Find:

Number of vacancies per cubic meter at a given temperature.

b. Given:

$T = 850 \text{ degrees C} = 1123 \text{ K}$

$Q_v = 1.08 \text{ eV/atom}$

Density of Fe (ρ_{Fe}) = 7.65 g/cc

Atomic weight of iron (A_{Fe}) = 7.65 g/mol

c. Assumptions:

(i) The question asks for the equilibrium number of vacancies.

(ii) The system is always in equilibrium with its surroundings during the process of attaining the said temperature of 1123 K. (Consider a material 'quenched', i.e., rapidly cooled from a 'high' temperature of say, 1163K to 1123 K. Since the system has no time to attain thermal equilibrium, it will retain the vacancies that it had at 1163 K at 1123 K!)

d. Solution:

$$N_v = N \exp\left(-\frac{Q_v}{kT}\right) \quad \dots(1)$$

Step 1: Calculate N using one of two equivalent methods.

Method 1

From density data: 7.65 g $\equiv$ 1cc

$$\Rightarrow 1\text{g} \equiv (1/7.65) \text{ cc} \quad \dots(2)$$

From atomic weight data: 55.85 g $\equiv$ 1 mol

$$\Rightarrow 55.85 \text{ g} \equiv 6.023 * 10^{23} \text{ atoms}$$

$$\Rightarrow 1\text{g} \equiv 1/(6.023 * 10^{23}) \text{ atoms} \quad \dots(3)$$

Hence, from (2) and (3), $6.023 \times 10^{23} / 55.85 \text{ atoms} \equiv (1/7.65) \text{ cc}$

Or, there are $7.65 \times 6.023 \times 10^{23} / 55.85 \text{ atoms /cc}$.

Method 2

Alternately, the following formula may be used to directly calculate $N = \frac{N_A \rho_{Fe}}{A_{Fe}}$

And,

$$N_v = \frac{N_A \rho_{Fe}}{A_{Fe}} \exp\left(-\frac{Q_v}{kT}\right)$$

Step 2: Substitute values in equation (1).

$$N_v = \frac{(6.022 \times 10^{23} \text{ atoms/mol})(7.65 \text{ g/cm}^3)}{55.85 \text{ g/mol}} \exp\left[-\frac{1.08 \text{ eV/atom}}{(8.62 \times 10^{-5} \text{ eV/atom-K})(850^\circ\text{C} + 273 \text{ K})}\right]$$

$$\mathbf{N_v = 1.18 \times 10^{18} \text{ cm}^{-3}}$$

OR

$$\mathbf{N_v = 1.18 \times 10^{24} \text{ m}^{-3}}$$

Problem 4.9

a. To Find:

The weight percentages of the constituents of an alloy, given the weights of these individual constituents.

b. Given:

Mass of titanium, $m_{Ti} = 218$ kg

Mass of aluminum, $m_{Al} = 14.6$ kg

Mass of vanadium, $m_V = 9.7$ kg

c. Assumptions:

The alloy contains no other alloying agents – the presence of other elements will modify the values of the weight percentages.

d. Solution:

For this alloy, the concentration of titanium (C_{Ti}) :

$$\begin{aligned} C_{Ti} &= \frac{m_{Ti}}{m_{Ti} + m_{Al} + m_V} \times 100 \\ &= \frac{218 \text{ kg}}{218 \text{ kg} + 14.6 \text{ kg} + 9.7 \text{ kg}} \times 100 = 89.97 \text{ wt\%} \end{aligned}$$

Similarly, for the concentration of aluminum (C_{Al}) :

$$C_{Al} = \frac{14.6 \text{ kg}}{218 \text{ kg} + 14.6 \text{ kg} + 9.7 \text{ kg}} \times 100 = 6.03 \text{ wt\%}$$

For the concentration of vanadium (C_V) :

$$C_V = \frac{9.7 \text{ kg}}{218 \text{ kg} + 14.6 \text{ kg} + 9.7 \text{ kg}} \times 100 = 4.00 \text{ wt\%}$$

$$\boxed{C_{Ti} = 89.97 \text{ wt\%} ; C_{Al} = 6.03 \text{ wt\%}; C_V = 4 \text{ wt\%}}$$

Problem 4.11

a. To Find:

To find the composition, in atom percent, of an alloy, given the weights of the individual constituents of the alloy.

b. Given:

Mass of copper, $m_{\text{Cu}} = 99.7 \text{ lb}$

Mass of zinc, $m_{\text{Zn}} = 102 \text{ lb}$

Mass of lead, $m_{\text{Pb}} = 2.1 \text{ lb}$

c. Assumptions:

The alloy contains no other alloying agents – the presence of other elements will modify the composition.

d. Solution:

Step 1: Convert the masses from lb to g.

$$m_{\text{Cu}} = (99.7 \text{ lb}_m)(453.6 \text{ g/lb}_m) = 45,224 \text{ g}$$

$$m_{\text{Zn}} = (102 \text{ lb}_m)(453.6 \text{ g/lb}_m) = 46,267 \text{ g}$$

$$m_{\text{Pb}} = (2.1 \text{ lb}_m)(453.6 \text{ g/lb}_m) = 953 \text{ g}$$

Step 2: For each element, convert its individual mass to corresponding number of moles.

$$n_{m_{\text{Cu}}} = \frac{m_{\text{Cu}}}{A_{\text{Cu}}} = \frac{45,224 \text{ g}}{63.55 \text{ g/mol}} = 711.6 \text{ mol}$$

$$n_{m_{\text{Zn}}} = \frac{m_{\text{Zn}}}{A_{\text{Zn}}} = \frac{46,267 \text{ g}}{65.41 \text{ g/mol}} = 707.3 \text{ mol}$$

$$n_{m_{\text{Pb}}} = \frac{m_{\text{Pb}}}{A_{\text{Pb}}} = \frac{953 \text{ g}}{207.2 \text{ g/mol}} = 4.6 \text{ mol}$$

Note : Atom percent is the same as mole percent. It is possible to convert all the above values from moles to atoms by multiplying 711.6, 707.3 and 4.6 by 6.023×10^{23} . It is also possible to use these values in moles directly, since the factor of 6.023×10^{23} in the numerator and denominator will cancel out in the final step, Step 3.

Step 3: Calculation of atom % from number of moles of individual constituents

$$\begin{aligned}\text{Atom \% of a constituent} &= (\text{Number of moles of a constituent} / \text{Total number of moles}) \times 100 \\ &= (\text{Number of atoms of a constituent} / \text{Total number of atoms}) \times 100\end{aligned}$$

Atom % of Cu

$$\begin{aligned}C'_{\text{Cu}} &= \frac{n_{m_{\text{Cu}}}}{n_{m_{\text{Cu}}} + n_{m_{\text{Zn}}} + n_{m_{\text{Pb}}}} \times 100 \\ &= \frac{711.6 \text{ mol}}{711.6 \text{ mol} + 707.3 \text{ mol} + 4.6 \text{ mol}} \times 100 = 50.0 \text{ at\%}\end{aligned}$$

Atom % of Zn

$$\text{Similarly, } C'_{\text{Zn}} = \frac{707.3 \text{ mol}}{711.6 \text{ mol} + 707.3 \text{ mol} + 4.6 \text{ mol}} \times 100 = 49.7 \text{ at\%}$$

Atom % of Pb

$$\text{Also, } C'_{\text{Pb}} = \frac{4.6 \text{ mol}}{711.6 \text{ mol} + 707.3 \text{ mol} + 4.6 \text{ mol}} \times 100 = 0.3 \text{ at\%}$$

Cu: 50 at%; Zn: 49.7 at%, Pb: 0.3 at%

Problem 4.18

a. To Find:

Determine the crystal structure of the given hypothetical alloy / Determine 'n' for the given hypothetical alloy.

b. Given:

- (i) Cubic unit cell/ cubic system/ crystal structure is one of the following : simple cubic, face-centered cubic, body-centered cubic
- (ii) Composition of alloy : $C_A = 12.5 \text{ wt\% A}$, $C_B = 87.5 \text{ wt\% B}$
- (iii) Density of A (r_A) = 4.27 g/cc, Density of B (r_B) = 6.35 g/cc
- (iv) Atomic weight of A (A_A) = 61.4 g/mol , Atomic weight of B (B_B) = 125.7 g/mol
- (v) Lattice parameter / edge length (a) = 0.395 nm.

c. Assumptions:

- (i) The study is conducted at constant temperature (the same temperature at which the lattice parameter is measured.) If the temperature changes, the lattice parameter/ edge length of the unit cell may change.
- (ii) The hypothetical alloy is assumed to have no vacancies (or self-interstitials), line defects or grain-boundaries. To allow us to approximate the equilibrium number of vacancies as zero, the temperature at which the study is conducted must be very low.
- (iii) Density values given correspond to those for perfect crystals.
- (iv) If the calculated value of 'n' is not exactly a whole number, this will be assumed to arise from minor inaccuracies in the input data used to calculate 'n'.
- (v) Hard sphere model of atom.

d. Solution:

Equation 3.5 is modified in order to reflect average values for density and atomic weight.

$$\rho_{ave} = \frac{nA_{ave}}{V_C N_A} \quad \dots(1)$$

Step 1:

$$V_C = a^3 \quad \dots(2)$$

$\Rightarrow$

$$n = \frac{\rho_{ave} a^3 N_A}{A_{ave}} \quad \dots(3)$$

Step 2:

ρ_{ave} may be calculated using two methods.

Method 1

Direct substitution of values in Equation 4.10a –

$$\rho_{ave} = \left(\frac{100}{\frac{C_A}{\rho_A} + \frac{C_B}{\rho_B}} \right) \quad \dots(4)$$

Method 2

Let total mass be 100g. Then, mass of A = $C_A = 12.5\text{g}$ and mass of B = $C_B = 87.5\text{g}$.

Total Volume = Volume of A + Volume of B = (Mass of A/Density of A) + (Mass of B/Density of B)

$\rho_{ave} = \text{Total Mass} / \text{Total Volume}.$

This yields equation (4).

This method allows us to derive equation 4.10a !

Step 3:

A_{ave} may be calculated using two methods.

Method 1

Direct substitution of values in Equation 4.11a-

$$A_{ave} = \left(\frac{100}{\frac{C_A}{A_A} + \frac{C_B}{A_B}} \right) \dots(5)$$

Method 2

Again, let the total mass be 100g. Then, mass of A = $C_A = 12.5\text{g}$ and mass of B = $C_B = 87.5\text{g}$.

Number of moles of A (N_A) = C_A/A_A

Number of moles of B (N_B) = C_B/A_B

Mole fraction of A (X_A) = $N_A / (N_A + N_B)$

Mole fraction of B (X_B) = $N_B / (N_A + N_B)$

Then, $A_{ave} = (X_A * A_A) + (X_B * A_B)$

This yields equation (5).

This method allows us to derive equation 4.11a !

Step 4: Solving for n

$$n = \frac{\left(\frac{100}{\frac{C_A}{\rho_A} + \frac{C_B}{\rho_B}} \right) a^3 N_A}{\left(\frac{100}{\frac{C_A}{A_A} + \frac{C_B}{A_B}} \right)}$$

$$\Rightarrow n = \frac{\left(\frac{100}{\frac{12.5 \text{ wt}\%}{4.27 \text{ g/cm}^3} + \frac{87.5 \text{ wt}\%}{6.35 \text{ g/cm}^3}} \right) (3.95 \times 10^{-8} \text{ nm})^3 (6.022 \times 10^{23} \text{ atoms/mol})}{\left(\frac{100}{\frac{12.5 \text{ wt}\%}{61.4 \text{ g/mol}} + \frac{87.5 \text{ wt}\%}{125.7 \text{ g/mol}}} \right)}$$

$$\Rightarrow n \approx 2$$

n = 2
 $\Rightarrow$ **Crystal Structure : BCC**

Problem 4.D2

a. To Find:

Weight % of V in solid solution with Fe, given the lattice parameter/unit cell edge length.

b. Given:

- (i) V forms a substitutional solid solution with Fe. The maximum solid solubility of V in Fe at room temperature (i.e., the maximum amount of V that can dissolve in Fe at room temperature) is 20 wt %. Hence, our answer should be ≤ 20 wt %.
- (ii) The lattice parameter/ unit cell edge length of this alloy is 0.289 nm.
- (iii) This alloy has a bcc crystal structure. Hence, $n=2$.

c. Assumptions:

- (i) Hard sphere model of atom.
- (ii) The alloy is assumed to have no vacancies (or self-interstitials), line defects or grain-boundaries.
- (iii) The study is being conducted at room-temperature. The alloy is allowed to reach room-temperature slowly, post-processing, such that this solid-solution is the equilibrium phase. Alternately, the (bcc) crystal structure that this alloy has, is associated with the lowest free energy at room temperature and will not transform to a different structure.
- (iv) No other alloying agent is added.

d. Solution:

Equation 3.5 is modified in order to reflect average values for density and atomic weight.

$$V_C = \frac{nA_{\text{ave}}}{\rho_{\text{ave}}N_A} \quad \dots(1)$$

Step 1: Calculation of volume of unit cell.

$$V_C = a^3 = (0.289 \text{ nm})^3 = (2.89 \times 10^{-8} \text{ cm})^3 = 2.414 \times 10^{-23} \text{ cm}^3$$

Step 2: Calculation of A_{ave}

Method 1 (equation 4.11a)

$$A_{ave} = \frac{100}{\frac{C_V}{A_V} + \frac{(100 - C_V)}{A_{Fe}}} = \frac{100}{\frac{C_V}{50.94 \text{ g/mol}} + \frac{(100 - C_V)}{55.85 \text{ g/mol}}} \quad \dots(2)$$

Method 2

Let the total weight of the alloy be 100g. If the weight of V is ' C_V ' g, the weight of y must be ' $100-C_V$ ' g.

Number of moles of V, $N_V = x/50.94$

Number of moles of Fe, $N_{Fe} = (100-x)/55.84$

Mole fraction of V, $X_V = N_V/N_V+N_{Fe}$

Mole fraction of V, $X_{Fe} = N_{Fe}/N_V+N_{Fe}$

$$A_{ave} = (X_V * A_V) + (X_{Fe} * A_{Fe})$$

This allows us to arrive at equation (2). In essence, we have derived equation 4.11a.

Step 3: Calculation of ρ_{ave}

Method 1 (equation 4.10a)

$$\rho_{ave} = \frac{100}{\frac{C_V}{\rho_V} + \frac{(100 - C_V)}{\rho_{Fe}}} = \frac{100}{\frac{C_V}{6.10 \text{ g/cm}^3} + \frac{(100 - C_V)}{7.87 \text{ g/cm}^3}} \quad \dots(3)$$

Method 2

Let the total weight of the alloy be 100g. If the weight of V is ' C_V ' g, the weight of y must be ' $100-C_V$ ' g.

Total Volume = Volume of V + Volume of Fe = (Mass of V/Density of V) + (Mass of Fe/Density of Fe)

$$\rho_{\text{ave}} = \text{Total mass} / \text{Total Volume}$$

Thus, this method allows us to arrive at expression (3) by deriving 4.10a!

Step 4: Solving for C_V

From (1), (2) and (3),

$$2.414 \times 10^{-23} \text{ cm}^3 = \frac{(2 \text{ atoms/unit cell}) \left[\frac{100}{\frac{C_V}{50.94 \text{ g/mol}} + \frac{(100 - C_V)}{55.85 \text{ g/mol}}} \right]}{\left[\frac{100}{\frac{C_V}{6.10 \text{ g/cm}^3} + \frac{(100 - C_V)}{7.87 \text{ g/cm}^3}} \right]} (6.022 \times 10^{23} \text{ atoms/mol})$$

Composition of V, $C_V = 12.9 \text{ wt}\%$

Note 1: In the above problems , the terms ‘ mass’ and ‘weight’ have both been used. Both are used to indicate ‘mass’.

Note 2: If Method 2 has been employed to evaluate A_{ave} and ρ_{ave} in Problems 4.18 and 4.D2, it is possible that the expression obtained in Step 4 in these two problems may differ in appearance from that given in the solution. If the final answer is correct, it’s probably all correct.